CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-503: Computer Networks

Session: December 2016 Examination | **Paper Code:** 18023 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) **Which device is used for Modulation and Demodulation? Explain its function.**

Answer: A **MODEM (Modulator-Demodulator)**. Converts digital computer pulses to analog signals for transmission over analog lines (modulation) and translates received analog waves back into digital bits (demodulation).

Q2 (3M) **What is Network Topology? List advantages and disadvantages of Mesh Topology.**

Mesh Topology:

- *Advantages:* Dedicated point-to-point links eliminate traffic congestion; robust fault tolerance (link failure affects only two nodes); excellent privacy.
- *Disadvantages:* Requires massive cabling (N(N-1)/2 links) and expensive multi-port hardware; complex installation.

Q3 (3M) **Differentiate between Simplex, Half-Duplex, and Full-Duplex Transmission.**

Differences: Simplex is unidirectional, Half-duplex is alternate bidirectional, Full-duplex is simultaneous bidirectional.

Q4 (3M) **What is Bandwidth of a channel?**

Definition: The range of frequencies contained in a signal or passed by a transmission channel: B = f_{max} - f_{min} (measured in Hertz, or bps in digital communication).

Q5 (3M) **What is a Protocol in Computer Networks?**

Definition: A formal set of rules and conventions governing the format, timing, sequencing, and error control of data transmission between communicating entities (Syntax, Semantics, Timing).

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) **Explain Seven Layers of ISO/OSI Reference Model.**

OSI Model: Physical, Data Link, Network, Transport, Session, Presentation, Application layers with primary duties.

Q7 (7.5M) **Explain Sliding Window Protocol: Stop-and-Wait vs Sliding Window.**

Comparison: Stop-and-Wait sends 1 frame at a time; Sliding Window allows sending W frames before receiving acknowledgments, multiplying channel utilization.

Q8 (7.5M) **Explain Distance Vector Routing Algorithm with Bellman-Ford equation.**

Algorithm: Path updating using D_x(y) = \min_v \{ c(x, v) + D_v(y) \} across neighboring nodes.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Explain Error Control using CRC with an example bit sequence.**

CRC Division: Modulo-2 binary arithmetic and frame checksum verification.

Q10 (15M) **Explain Guided Media: Twisted Pair, Coaxial Cable, and Optical Fiber.**

Guided Media: Full analysis of transmission mechanisms, attenuation, and physical connectors.

Q11 (15M) **Explain Switching Techniques: Circuit Switching, Message Switching, and Packet Switching.**

Switching Methods: Architectural breakdown, delay calculations, and applications.

Q12 (15M) **Explain Congestion Control: Leaky Bucket and Token Bucket Algorithms.**

Traffic Shaping: Detailed algorithms, diagrams, and burst handling.

Q13 (15M) **Explain IP Addressing (IPv4 Classes A, B, C, D, E) and Subnetting.**

IP Addressing: Address format, network IDs, host IDs, default masks, and subnet mask calculation.

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-503: Computer Networks

Session: December 2017 Examination | **Paper Code:** 18023 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) Repeater operates in which layer of the OSI model? Explain its function.

Explanation: A **Repeater** operates at the **Physical Layer (Layer 1)** of the OSI model. It receives weak or distorted electrical/optical signals from one cable segment, cleans, amplifies, regenerates the signal to its original strength and retimes it before transmitting it over the next segment, overcoming transmission attenuation.

Q2 (3M) Differentiate between FDM and TDM Multiplexing.

Differences: FDM divides available channel bandwidth into distinct non-overlapping frequency bands (analog). TDM allocates the entire channel bandwidth to each source for dedicated repeating time slots (digital).

Q3 (3M) A local telephone network is an example of which switching technology?

Answer: Circuit Switching. A dedicated physical communication circuit is established between the calling and called parties through telephone exchanges for the entire duration of the call.

Q4 (3M) What is Guided Media? List examples.

Definition: Physical solid transmission media that provide a physical conduit for electromagnetic signals: Twisted Pair Cable, Coaxial Cable, and Fiber Optic Cable.

Q5 (3M) Compare TCP and UDP at the Transport Layer.

Comparison: TCP is connection-oriented, reliable (acknowledgments, retransmissions), with flow and congestion control (HTTP, FTP). UDP is connectionless, unreliable, lightweight, and low-latency (DNS, VoIP, video streaming).

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) Explain Seven Layers of ISO/OSI Reference Model.

OSI Architecture: Physical, Data Link, Network, Transport, Session, Presentation, Application layers with primary duties.

Q7 (7.5M) Explain Stop-and-Wait Flow Control Protocol and calculate its efficiency.

Efficiency: \eta = \frac{T_t}{T_t + 2T_p} = \frac{1}{1 + 2a} where a = \frac{\text{Propagation Time}}{\text{Transmission Time}}.

Q8 (7.5M) Explain Dijkstra's Algorithm for Shortest Path Routing.

Dijkstra's Algorithm: Iteratively selects minimum-cost unvisited nodes to build the shortest-path routing tree.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) Explain Error Control using CRC with complete mathematical division example.

CRC Division: Modulo-2 polynomial division with verification at receiver side.

Q10 (15M) **Explain Sliding Window Protocols: Go-Back-N and Selective Repeat.**

Sliding Window: In-depth operation, sliding window mechanics, and timeout handling.

Q11 (15M) **Explain Transmission Media: UTP, STP, Coaxial, and Optical Fiber.**

Detailed Media: Physical construction, connectors (RJ45, BNC, SC/ST), bandwidth, and propagation speed.

Q12 (15M) **Explain Congestion Control Algorithms: Leaky Bucket and Token Bucket.**

Traffic Shaping: Comparison of uniform leak rate vs burst absorption using tokens.

Q13 (15M) **Explain Switching Techniques: Circuit, Packet, and Message Switching with detailed diagrams.**

Switching Architectures: Latency, store-and-forward delay, and buffer utilization.

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-503: Computer Networks

Session: December 2018 Examination | **Paper Code:** 18023 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) What is the difference between TCP and OSI Model?

Differences: OSI is a 7-layer theoretical conceptual model developed by ISO. TCP/IP is a 4-layer practical protocol suite developed by DoD that forms the basis of the modern Internet. OSI strictly separates session and presentation layers, whereas TCP/IP combines them into the Application layer.

Q2 (3M) What is Network Topology? Mention any two.

Topologies: Star topology (all nodes connect to a central hub/switch) and Bus topology (all devices share a single backbone cable terminated at both ends).

Q3 (3M) Explain Transmission Modes: Simplex, Half-Duplex, and Full-Duplex.

Modes: Simplex is strictly one-way (keyboard to monitor). Half-Duplex is bidirectional but one direction at a time (walkie-talkie). Full-Duplex is simultaneous two-way transmission (telephone conversation).

Q4 (3M) What is Modulation?

Definition: The process of varying one or more properties (amplitude, frequency, phase) of a high-frequency carrier wave with a data-bearing modulating signal for long-distance transmission.

Q5 (3M) What is Bit Rate vs Baud Rate?

Definitions: Bit rate is the number of bits transmitted per second (bps). Baud rate is the number of signal unit changes (symbols) per second. \text{Bit Rate} = \text{Baud Rate} \times \text{Bits per Baud}.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) Explain any one protocol used for flow control in noisy channel: Stop-and-Wait ARQ.

Stop-and-Wait ARQ: Sender sends frame with sequence number (0 or 1) and starts timer. If positive ACK received, next frame is sent. If ACK lost or timer expires, frame is retransmitted. Handles lost frames and duplicate frames using 1-bit sequence numbering.

Q7 (7.5M) Explain Fiber Distributed Data Interface (FDDI) operation.

FDDI: High-speed 100 Mbps token-passing optical fiber network based on dual counter-rotating token rings (Primary and Secondary). If the primary ring breaks, the dual ring self-heals by wrapping around, ensuring fault-tolerant operation.

Q8 (7.5M) Explain Distance Vector Routing Algorithm.

Explanation: Routers periodically advertise cost estimates to all known destinations to immediate neighbors. Uses Bellman-Ford algorithm to compute least-cost paths.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Explain ISO/OSI Reference Model in detail with all seven layers and functions.**

OSI Model: Complete 7-layer functional architecture, services, and protocols.

Q10 (15M) **Explain Sliding Window Protocols: Go-Back-N and Selective Repeat with comparisons.**

Protocols: Window sizing, ACK mechanism (cumulative vs individual), buffering requirements, and network utilization efficiency.

Q11 (15M) **Explain Error Detection using CRC (Cyclic Redundancy Check) with complete mathematical example.**

CRC Algorithm: Bit addition, modulo-2 division, remainder computation, and receiver verification.

Q12 (15M) **Explain Guided Media: Twisted Pair, Coaxial Cable, and Optical Fiber with advantages and limitations.**

Guided Media: Full physical characteristics, bandwidth capacity, attenuation rates, and shielding.

Q13 (15M) **Explain Circuit Switching, Packet Switching, and Message Switching with detailed comparison.**

Switching Methods: Complete comparison of connection setup, transmission delays, and resource utilization.

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-503: Computer Networks

Session: November 2019 Examination | **Paper Code:** 18023 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) Differentiate between Bridges and Repeaters.

Differences: Repeaters operate at the Physical Layer (Layer 1) to regenerate electrical signals without looking at frames. Bridges operate at the Data Link Layer (Layer 2) to inspect MAC addresses and filter traffic between segments.

Q2 (3M) What are the services provided by the Transport Layer?

Services: Connection-oriented delivery (TCP) vs connectionless datagram delivery (UDP), port-based process addressing, segmentation/reassembly, end-to-end flow control, and error recovery.

Q3 (3M) Define Modulation and Demodulation.

Explanation: Modulation transforms digital bitstreams into analog signals for transmission across physical media. Demodulation recovers original digital bits from incoming modulated analog waves.

Q4 (3M) What is Network Topology? Name two topologies with advantages.

Topologies: Star topology (central hub, easy fault isolation) and Mesh topology (point-to-point dedicated links, highest fault tolerance).

Q5 (3M) What is Shannon Capacity Theorem?

Theorem: C = B \log_2(1 + SNR) gives the theoretical upper bound on data transmission speed in bits per second over a noisy channel.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) Explain Dijkstra's Algorithm with an example network.

Dijkstra's Algorithm: Finds single-source shortest path to all destinations by maintaining tentative distance labels and iteratively selecting the minimum-cost unvisited node.

Q7 (7.5M) Explain Distance Vector Routing and the Count-to-Infinity problem.

Distance Vector Routing: Uses Bellman-Ford algorithm; slow reaction to link failures causes loops (count-to-infinity), resolved via split horizon.

Q8 (7.5M) Explain Sliding Window Protocol for Flow Control.

Sliding Window: Permits sending multiple frames within a window without waiting for immediate individual acknowledgments.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) Explain ISO/OSI Reference Model in detail with all seven layers.

OSI Architecture: Full 7-layer functional breakdown and data encapsulation process.

Q10 (15M) **Explain Guided Media: Twisted Pair Cable, Coaxial Cable, and Fiber Optic Cable.**

Transmission Media: Construction, signal propagation, attenuation characteristics, and network cabling standards (Cat5e, Cat6, Single-mode/Multi-mode fiber).

Q11 (15M) **Explain Error Detection using CRC (Cyclic Redundancy Check) with numerical problem.**

CRC Problem: Generator polynomial division, bit stuffing, and frame validation on receiver side.

Q12 (15M) **Explain Circuit Switching, Packet Switching, and Message Switching with comparison.**

Switching Techniques: Comprehensive comparison of latency, bandwidth efficiency, and modern telecom implementations.

Q13 (15M) **Explain Congestion Control in Computer Networks: Leaky Bucket and Token Bucket Algorithms.**

Traffic Shaping: Deep dive into burst mitigation, token accumulation, and packet scheduling.

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-503: Computer Networks

Session: December 2020 Examination | **Paper Code:** 18023 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) Four channels, each with 100-KHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 KHz between channels?

Solution: N = 4 channels of 100 KHz each.
Number of guard bands required between 4 channels = 4 - 1 = 3.
Each guard band = 10 KHz.
Total minimum bandwidth:
B = (4 \times 100) + (3 \times 10) = 400 + 30 = \mathbf{430 \text{ KHz}}

Q2 (3M) Discuss Point-to-Point Protocol (PPP).

PPP: A data link layer protocol used to establish a direct connection between two nodes over dial-up, serial cable, or broadband (PPPoE). Provides authentication via PAP/CHAP, framing, and dynamic IP configuration (IPCP).

Q3 (3M) Explain the functions of the Presentation Layer.

Functions: Data syntax translation, character code conversion, data compression (lossless/lossy), and data encryption/decryption (cryptography).

Q4 (3M) Differentiate between LAN, MAN, and WAN.

Comparison: LAN covers a room or building (high speed, private ownership). MAN covers a city (cable TV network). WAN covers a country or globe (telecom lines, satellite, Internet).

Q5 (3M) What is a Collision Domain vs Broadcast Domain?

Definitions: A Collision Domain is a network segment where simultaneous transmissions collide (divided by switches). A Broadcast Domain is a segment receiving all broadcast frames (divided by routers).

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) How does DTE-DCE interface work? Discuss the essence of Modems.

DTE-DCE Interface: DTE (Data Terminal Equipment, e.g. computer/router) generates digital data. DCE (Data Circuit-terminating Equipment, e.g. modem) converts signals for transmission across the physical telephone link. Standards include RS-232 / EIA-232.

Q7 (7.5M) Explain Error Detection using CRC with generator polynomial $x^3 + x + 1$.

CRC Calculation: Complete modulo-2 polynomial division with bit representation `1011`.

Q8 (7.5M) Explain Distance Vector Routing Algorithm.

Algorithm: Periodic exchange of routing tables between adjacent neighbors based on Bellman-Ford algorithm.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Explain OSI Reference Model with detailed layer descriptions.**

7-Layer Model: Detailed functions, protocols, and data units (Bits, Frames, Packets, Segments, Data).

Q10 (15M) **Explain Sliding Window Protocols: Stop-and-Wait, Go-Back-N, and Selective Repeat.**

Flow Control: Efficiency equations, window buffering, and timeout retransmission mechanisms.

Q11 (15M) **Explain Transmission Media: UTP, STP, Coaxial, and Fiber Optic Cable.**

Transmission Media: Attenuation, propagation velocity, bandwidth, and EMI immunity analysis.

Q12 (15M) **Explain Congestion Control Algorithms: Leaky Bucket and Token Bucket.**

Traffic Shaping: Smooth constant-rate leakage vs bursty-tolerant token generation.

Q13 (15M) **Explain Switching Techniques: Circuit, Message, and Packet Switching.**

Switching: Detailed comparison of setup time, propagation delay, transmission delay, and queuing overhead.

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-503: Computer Networks

Session: December 2021 Examination | **Paper Code:** 18023 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks / CV-3 Pattern)

Q1 (7.5M) **Explain the functions of the Session Layer in the OSI model.**

Session Layer (Layer 5): Responsible for establishing, maintaining, synchronizing, and terminating communication sessions between applications. Controls dialog (simplex/half-duplex/full-duplex) and inserts synchronization checkpoints into data streams to facilitate restart upon failure.

Q2 (7.5M) **Write short notes on: (a) Throughput & Wavelength, (b) Synchronous Protocols, (c) Routers.**

Notes:

- **(a) Throughput & Wavelength:** Throughput is the actual measurement of data transferred across a network per second. Wavelength (\lambda = v / f) is the physical distance between consecutive signal wave peaks in optical communication.
- **(b) Synchronous Protocols:** Continuous transmission of data blocks synchronized to a shared clock, using sync bytes/flags (e.g. HDLC).
- **(c) Routers:** Network Layer (Layer 3) internetworking devices that inspect destination IP addresses and make packet forwarding decisions based on routing tables.

Q3 (7.5M) **Discuss ISDN, its services, and layer structure.**

ISDN (Integrated Services Digital Network): Transmits voice, data, and video simultaneously over standard telephone copper lines. Channels: B-channel (64 kbps, user data), D-channel (16 or 64 kbps, control signaling). BRI provides 2B+D (144 kbps); PRI provides 23B+D (1.544 Mbps) or 30B+D (2.048 Mbps).

Q4 (7.5M) **Explain Flow Control and Error Control in Computer Networks.**

Mechanisms: Flow control ensures a fast sender does not overwhelm a slow receiver (Sliding Window). Error control detects corrupted frames via CRC and retransmits them via ARQ schemes (Stop-and-Wait, Go-Back-N, Selective Repeat).

Q5 (7.5M) **Explain TCP Packet Format (Header Structure) in detail.**

TCP Header Fields:

- Source Port (16 bits) and Destination Port (16 bits)
- Sequence Number (32 bits) and Acknowledgment Number (32 bits)
- Header Length / Data Offset (4 bits) and Reserved bits (6 bits)
- Control Flags (6 bits: URG, ACK, PSH, RST, SYN, FIN)
- Window Size (16 bits) for flow control
- Checksum (16 bits) and Urgent Pointer (16 bits)

SECTION B — SHORT ANSWER QUESTIONS (1 × 15 = 15 Marks / CV-3 Pattern)

Q6 (15M) **Explain Seven Layers of ISO/OSI Reference Model with their functions.**

Complete OSI Model: Detailed explanation of Physical, Data Link, Network, Transport, Session, Presentation, and Application layers.

SECTION C — DETAILED ANSWER QUESTIONS (2 × 22.5 = 45 Marks / CV-3 Pattern)

Q7 (22.5M) **Explain Routing Algorithms: Shortest Path Routing (Dijkstra) and Distance Vector Routing with Count-to-Infinity problem.**

Routing Algorithms: Dijkstra link-state calculation vs Bellman-Ford distributed vector routing with split horizon and poison reverse solutions.

Q8 (22.5M) **Explain Sliding Window Protocols: Go-Back-N and Selective Repeat with window sizes and frame sequences.**

Sliding Window: Mathematical condition for maximum window sizes: Go-Back-N ($W_s \le 2^m - 1$) and Selective Repeat ($W_s + W_r \le 2^m \implies W_s \le 2^{m-1}$) to prevent ambiguity with duplicate sequence numbers.

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-503: Computer Networks

Session: December 2023 Examination | **Paper Code:** 18023 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) Explain the use of Bridges in Computer Networks.

Definition: A **Bridge** is a Data Link Layer (Layer 2) device that connects two separate LAN segments running the same protocol, inspecting MAC addresses to filter frames and segment collision domains.

Q2 (3M) Write a short note on E-Mail and its core protocols.

Explanation: Electronic mail architecture consists of User Agents (UAs) and Message Transfer Agents (MTAs). Uses SMTP for sending emails, and POP3 / IMAP for downloading emails to client devices.

Q3 (3M) What is Network Topology? Name major topologies.

Topologies: The geometric layout of interconnected nodes: Star, Bus, Ring, Mesh, Tree, and Hybrid.

Q4 (3M) What is Attenuation in transmission media?

Definition: The progressive loss of signal energy/strength as it propagates through transmission media, measured in decibels (dB), compensated using amplifiers or repeaters.

Q5 (3M) Define Bandwidth and Throughput.

Definitions:

- **Bandwidth:** The maximum theoretical data-carrying capacity of a communication channel (bits per second).
- **Throughput:** The actual rate of successfully transmitted data delivered across the network per unit time.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) Describe the functions & responsibilities of the Session Layer.

Session Layer (Layer 5):

- **Session Establishment, Maintenance, and Termination:** Manages dialogs between communicating processes.
- **Dialog Control:** Determines whether communication is simplex, half-duplex, or full-duplex.
- **Synchronization / Checkpointing:** Inserts sync tokens into data streams to allow recovery from checkpoints in case of interruptions.

Q7 (7.5M) **Differentiate between OSI Model and TCP/IP Model.**

Comparison:

Aspect	OSI Reference Model	TCP/IP Model
Layers	7 layers	4 layers
Origin	Theoretical standard created by ISO.	Practical protocol suite developed for ARPANET.
Session & Presentation	Separate distinct layers.	Integrated into Application layer.
Protocols	Protocols developed after model.	Model developed based on existing protocols.

Q8 (7.5M) **Explain IP Addressing and Subnetting.**

IP Addressing: Classes A, B, C, D, E. Subnetting borrows host bits to divide large networks into smaller subnets.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **What is Congestion Control? How is Congestion Control different from Flow Control? Explain various congestion control techniques.**

Congestion vs Flow Control:

- **Flow Control:** Point-to-point mechanism preventing a fast sender from overwhelming a single slow receiver (Sliding Window).
- **Congestion Control:** Network-wide mechanism preventing global traffic load from exceeding overall subnet router buffer capacity.

Techniques: Leaky Bucket algorithm, Token Bucket algorithm, Choke Packets, and Random Early Detection (RED).

Q10 (15M) **Explain Error Control in Data Link Layer using CRC with a complete numerical example.**

CRC Numerical Example:
Data: `1010000`, Generator: `1001`.
Append 3 zeros: `1010000000`. Perform modulo-2 binary division by `1001` to calculate remainder (FCS). Complete step-by-step arithmetic shown.

Q11 (15M) **Explain Dijkstra's Shortest Path Routing Algorithm with a network graph example.**

Dijkstra's Algorithm: Greedy shortest path calculation from source router to all other nodes maintaining visited sets and updating tentative distances.

Q12 (15M) **Explain Transmission Media: Twisted Pair, Coaxial Cable, and Optical Fiber.**

Guided Media: Bandwidth, attenuation, electromagnetic immunity, and cost trade-offs across all three physical media.

Q13 (15M) **Write short notes on: (a) Firewalls, (b) FTP, (c) Domain Name System (DNS).**

Detailed Notes: Packet filtering firewalls, dual-port FTP architecture (Control port 21, Data port 20), and hierarchical DNS domain resolution.

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-503: Computer Networks

Session: December 2024 Examination | **Paper Code:** 18023 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) **What is the need for a Computer Network?**

Need for Computer Networks: Resource sharing (printers, storage servers), centralized data management, cost reduction, electronic communication (email, video conferences), and distributed processing power.

Q2 (3M) **Define HDLC (High-Level Data Link Control).**

Definition: HDLC is a bit-oriented synchronous data link layer protocol developed by ISO that provides reliable point-to-point and multipoint communication using frame delimitation (flag byte `01111110` with bit stuffing), error detection, and sliding window flow control.

Q3 (3M) **Explain the concept of Piggybacking.**

Definition: Piggybacking is a flow control optimization technique where an acknowledgment (ACK) for received data is delayed and attached into the header of the next outgoing data frame sent back to the sender, eliminating the overhead of transmitting independent acknowledgment packets.

Q4 (3M) **Compare Datagram Approach and Virtual Circuit Approach in Packet Switching.**

Comparison:

Aspect	Datagram Approach (Connectionless)	Virtual Circuit Approach (Connection-Oriented)
Call Setup	None; packets transmitted independently.	Setup phase required before transmission.
Path	Each packet routed independently; may arrive out of order.	All packets follow identical pre-configured path in sequence.
Header Overhead	Large (full destination IP on every packet).	Small (short Virtual Circuit Identifier - VCI).

Q5 (3M) **Describe how SMTP protocol is used in E-Mail applications.**

SMTP (Simple Mail Transfer Protocol): A push protocol running over TCP port 25 that transfers email messages from a sender's user agent/client to the sender's mail server, and from the sender's mail server to the recipient's mail server across the Internet.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) **Discuss the functions of Transport, Presentation, and Application layers in the OSI model.**

Functions:

- **Transport Layer:** End-to-end segmentation, port addressing, connection management, reliable delivery, and flow control.
- **Presentation Layer:** Data translation (syntax formatting), encryption/decryption, and data compression.
- **Application Layer:** Direct user interfaces for network services (HTTP, FTP, SMTP, DNS).

Q7 (7.5M) **Explain the switching methods in Computer Networks.**

Switching Methods: Circuit Switching (dedicated physical communication path established before data transfer), Packet Switching (data split into discrete packets routed dynamically), and Message Switching (entire message stored and forwarded hop-by-hop).

Q8 (7.5M) **Explain in brief: (a) Guided Media, (b) Transmission Modes.**

(a) Guided Media: Physical physical conductors that guide signals along a specific path: Twisted Pair (UTP/STP), Coaxial Cable, and Optical Fiber.
(b) Transmission Modes:

- *Simplex:* Unidirectional transmission (e.g., keyboard to CPU, TV broadcast).
- *Half-Duplex:* Bidirectional, but only one direction at a time (e.g., Walkie-Talkie).
- *Full-Duplex:* Simultaneous bidirectional transmission (e.g., telephone call).

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **What are the layers of the ISO/OSI Protocol Stack? Briefly list their functions.**

Complete OSI Architecture: Physical, Data Link, Network, Transport, Session, Presentation, and Application layers with detailed protocol functions.

Q10 (15M) **Explain in detail about the Error & Flow Control mechanisms employed at the Data Link Layer.**

Mechanisms: Framing methods (byte stuffing, bit stuffing), error detection (CRC-32), and flow control ARQ schemes (Stop-and-Wait, Go-Back-N, Selective Repeat).

Q11 (15M) **(a) Explain Shannon Capacity with examples. (b) What is CSMA? List protocols used with CSMA.**

(a) Shannon Capacity Theorem: Gives theoretical maximum data rate C of a channel with bandwidth B and Signal-to-Noise Ratio SNR:
$C = B \log_2(1 + SNR) \text{ bps}$
Example: For telephone line with B = 3000 Hz and SNR = 3162 (35 dB): $C = 3000 \log_2(1 + 3162) \approx 3000 \times 11.62 = 34,860 \text{ bps} \approx 34.8 \text{ kbps}$.

(b) CSMA Protocols: 1-Persistent CSMA, Non-Persistent CSMA, p-Persistent CSMA, and CSMA/CD / CSMA/CA.

Q12 (15M) **(a) Explain ISDN Architecture and Services. (b) Describe Congestion Control Algorithms.**

(a) Integrated Services Digital Network (ISDN): Transmits voice, video, and data over digital telephone networks. BRI (2B+D = 144 kbps) and PRI (23B+D / 30B+D = 1.544 / 2.048 Mbps).
(b) Congestion Control: Open-loop (prevention via traffic policing) vs Closed-loop (reaction via choke packets and Leaky/Token Bucket algorithms).

Q13 (15M) **Describe any three: (a) SSL, (b) Modems, (c) Line Configuration, (d) Distortion vs Noise.**

Short Notes:

- **SSL (Secure Sockets Layer):** Cryptographic protocol providing end-to-end encryption, authentication, and integrity over TCP.
- **Modems (Modulator-Demodulator):** Converts digital computer signals to analog audio frequencies for transmission over analog lines and vice versa.
- **Line Configuration:** Point-to-Point (dedicated link between two devices) vs Multipoint (multidrop link where multiple devices share a single link).
- **Distortion vs Noise:** Distortion is alteration of signal shape caused by different frequency phase delays in transmission media. Noise is unwanted external thermal/electromagnetic energy added into the signal.

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-503: Computer Networks

Session: December 2025 Examination | **Paper Code:** 18023 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) What is a MAC address? How is it different from an IP Address?

Comparison:

Feature	MAC Address (Physical Address)	IP Address (Logical Address)
Layer	Data Link Layer (Layer 2).	Network Layer (Layer 3).
Size & Format	48 bits (6 bytes), hex (e.g., `00:1A:2B:3C:4D:5E`).	32 bits (IPv4, e.g., `192.168.1.1`) or 128 bits (IPv6).
Nature	Hardcoded by manufacturer into NIC; globally fixed.	Dynamically/statically assigned based on network location.

Q2 (3M) Differentiate between a Hub and a Switch.

Differences:

- **Hub (Physical Layer - Layer 1):** Broadcast device that repeats every incoming signal across all ports, sharing a single collision domain. Half-duplex, no intelligence.
- **Switch (Data Link Layer - Layer 2):** Intelligent device maintaining a MAC Address Table to forward frames directly to designated destination ports. Dedicated collision domains per port, full-duplex.

Q3 (3M) What is Subnetting? Give an example.

Definition: Subnetting is the process of partitioning a single physical network into two or more smaller logical subnets by borrowing host bits to create subnet bits, reducing broadcast traffic and optimizing address allocation.
Example: Given Class C network `192.168.1.0/24` with default mask `255.255.255.0`. Borrowing 2 host bits gives subnet mask `255.255.255.192` (/26), creating 2^2 = 4 subnets with 62 usable hosts each:

- Subnet 1: `192.168.1.0` to `192.168.1.63`
- Subnet 2: `192.168.1.64` to `192.168.1.127`
- Subnet 3: `192.168.1.128` to `192.168.1.191`
- Subnet 4: `192.168.1.192` to `192.168.1.255`

Q4 (3M) What is CSMA/CD? How does it operate in Ethernet?

Carrier Sense Multiple Access with Collision Detection: A media access protocol used in shared half-duplex Ethernet:

1. **Carrier Sense:** Station listens before transmitting ("listen before talk"). If channel is idle, it begins transmitting.
2. **Collision Detection:** While transmitting, station monitors signal level. If collision occurs, it transmits a *jam signal* to notify all stations.
3. **Backoff:** Stations wait a random backoff period using the Binary Exponential Backoff algorithm before retransmitting.

Q5 (3M) **Differentiate between Synchronous and Asynchronous Protocols.**

Comparison:

- **Asynchronous Transmission:** Characters sent independently with start (0) and stop (1) framing bits. Clocks do not need synchronization. Inefficient for large high-speed data transfers.
- **Synchronous Transmission:** Continuous stream of data blocks transmitted synchronized to an external shared clock. Synchronized using sync character preambles (e.g., HDLC, PPP), achieving high throughput.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) **Describe Multiplexing. Explain FDM, TDM, and WDM with diagrams.**

Definition: Combining multiple separate data streams into a single composite signal transmitted over a shared transmission medium.

Techniques:

- **FDM (Frequency Division Multiplexing):** Divides channel bandwidth into separate non-overlapping frequency bands with guard bands (Analog signals, radio, cable TV).
- **TDM (Time Division Multiplexing):** Interleaves time slots from multiple sources onto a high-speed digital carrier (Synchronous vs Statistical TDM).
- **WDM (Wavelength Division Multiplexing):** Multiplexes optical signals of differing wavelengths (colors) along a single optical fiber.

Q7 (7.5M) **Explain Error Detection Techniques: Parity Check, Checksum, and CRC (Cyclic Redundancy Check).**

Error Detection:

- **Parity Check:** Appends single parity bit (even or odd parity). Detects only odd numbers of bit errors.
- **Checksum:** Divides data into 16-bit words, adds them using 1's complement arithmetic, and appends inverted sum (used in TCP/IP).
- **Cyclic Redundancy Check (CRC):** Treats bit stream as polynomial coefficients and performs modulo-2 binary division by generator polynomial G(x). Appends remainder (FCS). Detects all single, double, odd, and burst errors up to generator length!

Q8 (7.5M) **Compare Circuit Switching, Packet Switching, and Message Switching.**

Comparison:

Feature	Circuit Switching	Packet Switching	Message Switching
Dedicated Path	Yes (setup before transfer).	No (packets routed dynamically).	No (store-and-forward per hop).
Bandwidth	Reserved/wasted if idle.	Dynamically shared on-demand.	Dynamically shared.
Delay	Setup delay high; transmission delay low.	Jitter/variable queuing delay.	High storage delay at each switch.
Modern Use	PSTN landline telephone.	Internet (IP networks).	Early telegraph, postal email.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Explain the ISO/OSI Reference Model in detail. Describe the primary functions of each of the seven layers.**

Seven Layers of OSI Model:

1. **Physical Layer (Layer 1):** Transmission of raw unstructured bit streams over physical media (voltages, connectors, cables).
2. **Data Link Layer (Layer 2):** Framing, physical addressing (MAC), flow control, error control (CRC), and media access control.
3. **Network Layer (Layer 3):** Logical addressing (IP), routing (Dijkstra, distance vector), packet forwarding, and fragmentation.
4. **Transport Layer (Layer 4):** End-to-end process-to-process delivery (ports), segmentation and reassembly, connection control (TCP handshake), reliable flow control.
5. **Session Layer (Layer 5):** Dialogue control (simplex, half/full duplex), session checkpointing, synchronization tokens.
6. **Presentation Layer (Layer 6):** Data translation, character code conversion (ASCII/EBCDIC), compression, encryption/decryption (SSL/TLS).
7. **Application Layer (Layer 7):** Network virtual terminals, file transfer (FTP), web browsing (HTTP), email services (SMTP).

Q10 (15M) **Explain Sliding Window Protocols: Stop-and-Wait ARQ, Go-Back-N ARQ, and Selective Repeat ARQ with efficiency analysis.**

Flow and Error Control Protocols:

- **Stop-and-Wait ARQ:** Sender transmits 1 frame, starts timer, and waits for ACK before sending next. Efficiency: \eta = \frac{1}{1 + 2a} where a = T_p / T_t. Poor for high bandwidth-delay links.
- **Go-Back-N ARQ:** Sender window size W_s = 2^m - 1, receiver window W_r = 1. Sender can transmit multiple frames before receiving ACK. On loss, retransmits lost frame and all subsequent frames. Cumulative ACKs.
- **Selective Repeat ARQ:** Sender window W_s = 2^{m-1}, receiver window W_r = 2^{m-1}. Receiver accepts out-of-order frames within window. Sender retransmits *only the specific corrupted/timed-out frames* (highest throughput!).

Q11 (15M) **Explain Routing Algorithms: (a) Distance Vector Routing (Bellman-Ford) and (b) Link State Routing (Dijkstra). Discuss Count-to-Infinity problem.**

Routing Algorithms:

1. **Distance Vector Routing:** Each router maintains a routing table containing distance to each destination and next hop. Routers periodically share complete tables with immediate neighbors. Uses Bellman-Ford algorithm. Vulnerable to *Count-to-Infinity problem* when links fail (mitigated by Split Horizon and Poison Reverse).
2. **Link State Routing:** Each router floods Link State Packets (LSP) containing status of its local links to all routers in the entire autonomous system. Every router constructs a global topology map and computes shortest paths using Dijkstra's algorithm (used in OSPF). Fast convergence and no routing loops.

Q12 (15M) **Explain Congestion Control Mechanisms in Computer Networks: Leaky Bucket and Token Bucket Algorithms with comparative analysis.**

Traffic Shaping Algorithms:

- **Leaky Bucket:** Discharges data at a strictly constant, uniform output rate regardless of incoming burstiness. If incoming burst exceeds bucket buffer capacity, excess packets overflow and drop. Smooths traffic completely.
- **Token Bucket:** Generates tokens at a constant rate up to bucket capacity. Allows transmission of bursty traffic as long as sufficient tokens exist, but limits sustained rate. More flexible and widely used in modern QoS.

Q13 (15M) **Write detailed notes on: (a) DNS (Domain Name System), (b) Firewalls and Network Security, (c) SMTP and Email Architecture.**

Detailed Notes:

- **DNS:** Hierarchical distributed database mapping hostnames (`www.google.com`) to IP addresses (`142.250.190.46`) over UDP/TCP port 53.
- **Firewalls:** Security barrier controlling inbound/outbound network traffic based on rules. Types: Packet Filtering, Stateful Inspection, Proxy/Application-level gateways.
- **Email Architecture:** User Agent (UA) composes email, Mail Transfer Agent (MTA) transmits using SMTP (port 25/587), and Mail Access Agent (MAA) retrieves email to client via POP3 (port 110) or IMAP (port 143).
